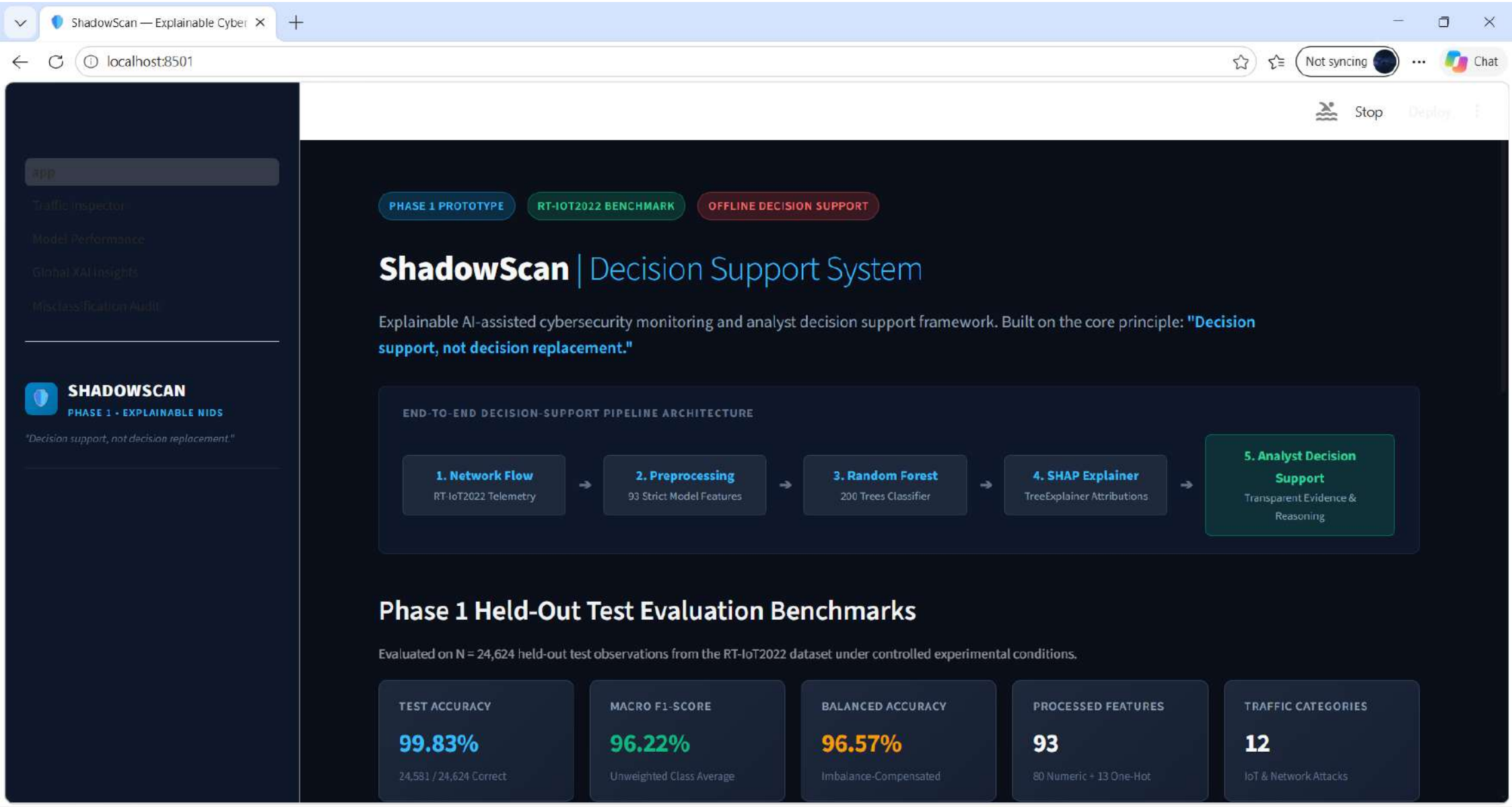

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Model Performance
Global XAI Insights
Misclassification Audit
SHADOWSCAN
PHASE 1 · EXPLAINABLE NIDS
"Decision support, not decision replacement."
PHASE 1 PROTOTYPE
RT-IOT2022 BENCHMARK
OFFLINE DECISION SUPPORT
ShadowScan | Decision Support System
Explainable AI-assisted cybersecurity monitoring and analyst decision support framework. Built on the core principle: "Decision support, not decision replacement."
END-TO-END DECISION-SUPPORT PIPELINE ARCHITECTURE
1. Network Flow
RT-IoT2022 Telemetry
2. Preprocessing
93 Strict Model Features
3. Random Forest
200 Trees Classifier
4. SHAP Explainer
TreeExplainer Attributions
5. Analyst Decision Support
Transparent Evidence & Reasoning
Phase 1 Held-Out Test Evaluation Benchmarks
Evaluated on N = 24,624 held-out test observations from the RT-IoT2022 dataset under controlled experimental conditions.
TEST ACCURACY
99.83%
24,581 / 24,624 Correct
MACRO F1-SCORE
96.22%
Unweighted Class Average
BALANCED ACCURACY
96.57%
Imbalance-Compensated
PROCESSED FEATURES
93
80 Numeric + 13 One-Hot
TRAFFIC CATEGORIES
12
IoT & Network Attacks
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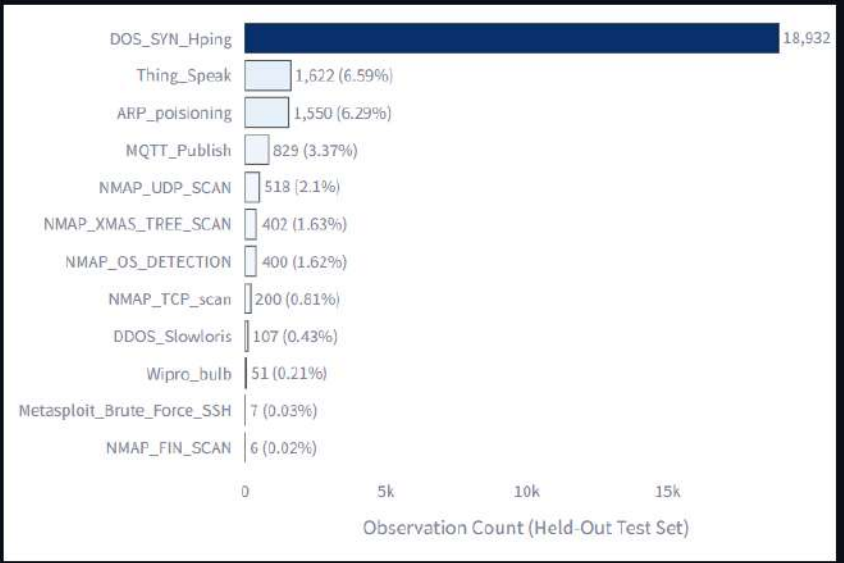
"Decision support, not decision replacement."

TEST ACCURACY	MACRO F1-SCORE	BALANCED ACCURACY	PROCESSED FEATURES	TRAFFIC CATEGORIES
99.83%	96.22%	96.57%	93	12
24,581 / 24,624 Correct	Unweighted Class Average	Imbalance-Compensated	80 Numeric + 13 One-Hot	IoT & Network Attacks

Technical Specifications

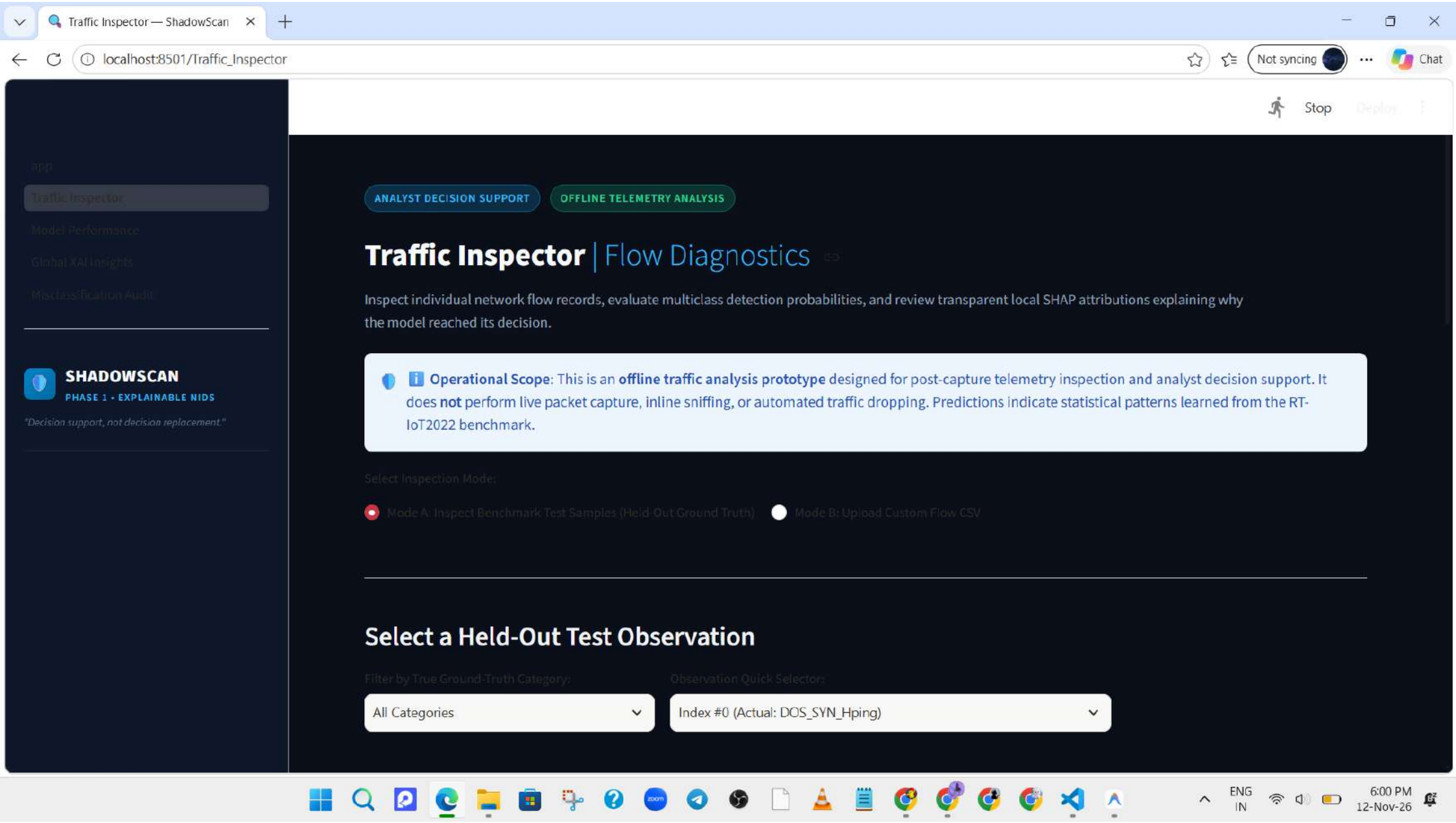
Specification	Detail
Source Dataset	RT-IoT2022 Network Traffic
Total Dataset Records	123,117 observations
Model-Training Partition	78,794 observations (80% of train)
Internal Validation Partition	19,699 observations (20% of train)
Held-Out Benchmark Test Partition	24,624 observations (stratified test)
Model Classifier	RandomForestClassifier (scikit-learn)
Number of Trees (Estimators)	200 decision trees (n_jobs=-1)
Model Random Seed	42 (reproducible seed)
Explainability Engine	SHAP TreeExplainer
Zero-Variance Dropped Feature	bwd_URG_flag_count (variance = 0)

Held-Out Test Class Distribution



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ANALYST DECISION SUPPORT
OFFLINE TELEMETRY ANALYSIS
Traffic Inspector | Flow Diagnostics
Inspect individual network flow records, evaluate multiclass detection probabilities, and review transparent local SHAP attributions explaining why the model reached its decision.
Operational Scope: This is an offline traffic analysis prototype designed for post-capture telemetry inspection and analyst decision support. It does not perform live packet capture, inline sniffing, or automated traffic dropping. Predictions indicate statistical patterns learned from the RT-IoT2022 benchmark.
Select Inspection Mode:
Mode A: Inspect Benchmark Test Samples (Held-Out Ground Truth)
Mode B: Upload Custom Flow CSV
Select a Held-Out Test Observation
Filter by True Ground-Truth Category:
All Categories
Observation Quick Selector:
Index #0 (Actual: DOS_SYN_Hping)
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Inspection & Decision Support: Test Flow Record #0

MODEL CLASSIFICATION	MODEL CONFIDENCE	GROUND TRUTH	PREDICTION VERIFICATION
DOS_SYN_Hping	100.00%	DOS_SYN_Hping	✓ Correct
Random Forest (200 Trees)	Top Class Probability	Benchmark Target	Model prediction matches ground truth.

› Full 12-Class Probability Distribution

Explainable Decision Support: Why was this decision made?

Classified as **DOS_SYN_Hping** with model confidence of **100.00%**.

The model's classification is mathematically driven by the feature contributions below. Features with **positive SHAP** increase model confidence for this category, while features with **negative SHAP** oppose it.

Key Drivers: Supporting vs. Opposing

Features Supporting the Classification (+SHAP):

- **Destination port** (`id.resp_p` = 21): pushed model attribution towards DOS_SYN_Hping (+0.0284 SHAP).

Interactive SHAP Attribution Breakdown



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Classified as **DOS_SYN_Hping** with model confidence of **100.00%**.

The model's classification is mathematically driven by the feature contributions below. Features with **positive SHAP** increase model confidence for this category, while features with **negative SHAP** oppose it.

Key Drivers: Supporting vs. Opposing

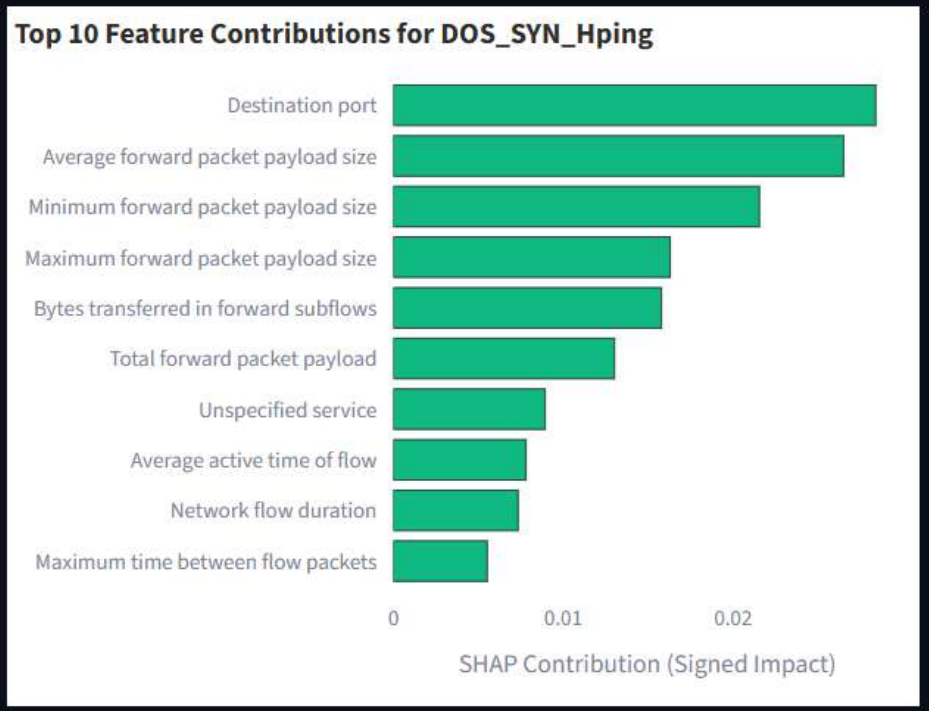
Features Supporting the Classification (+SHAP):

- **Destination port** (`id.resp_p` = 21): pushed model attribution towards DOS_SYN_Hping (+0.0284 SHAP).
- **Average forward packet payload size** (`fwd_pkts_payload.avg` = 120): pushed model attribution towards DOS_SYN_Hping (+0.0265 SHAP).
- **Minimum forward packet payload size** (`fwd_pkts_payload.min` = 120): pushed model attribution towards DOS_SYN_Hping (+0.0216 SHAP).
- **Maximum forward packet payload size** (`fwd_pkts_payload.max` = 120): pushed model attribution towards DOS_SYN_Hping (+0.0163 SHAP).
- **Bytes transferred in forward subflows** (`fwd_subflow_bytes` = 120): pushed model attribution towards DOS_SYN_Hping (+0.0158 SHAP).

Features Opposing the Classification (-SHAP):

- **id.orig** (

Interactive SHAP Attribution Breakdown



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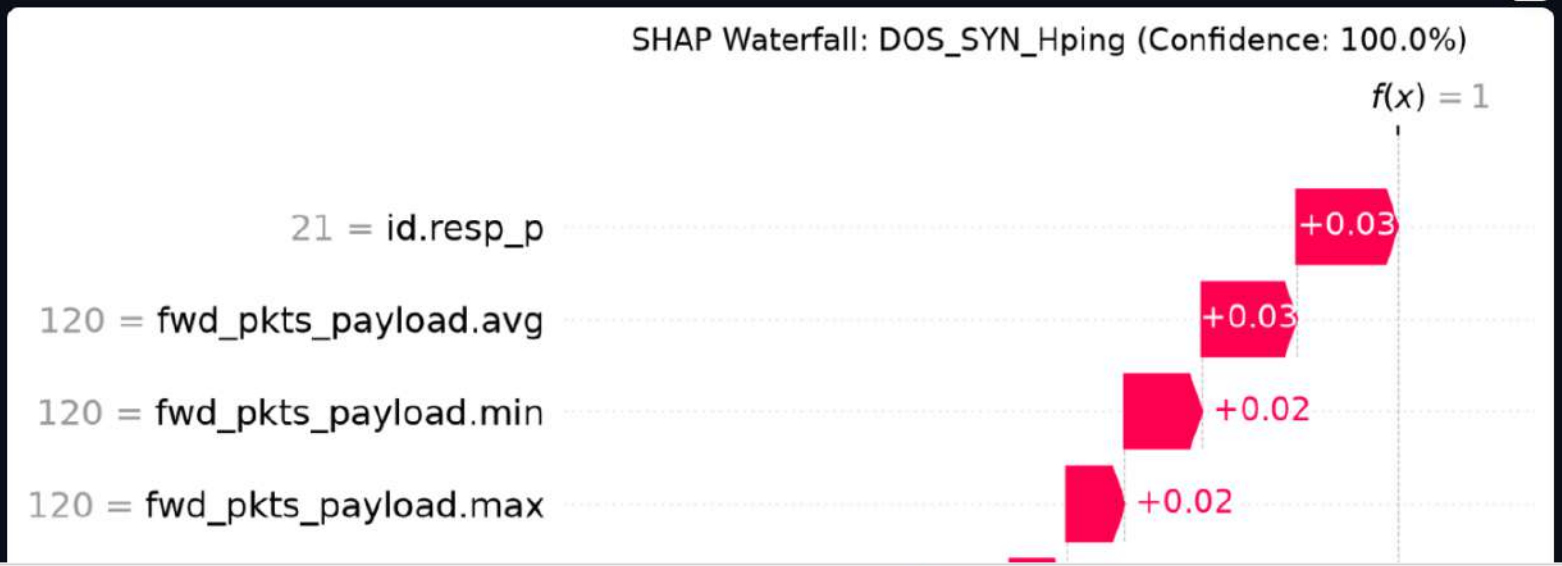
Features Opposing the Classification (-SHAP):

- **id.orig p** (`id.orig_p` = 5.499e+04): pushed model attribution away from DOS_SYN_Hping (-0.0003 SHAP).

📌 *Attributions are derived dynamically using SHAP TreeExplainer. Descriptions are mapped from the verified RT-IoT2022 domain glossary without fabricating security actions.*

Detailed SHAP Waterfall Plot

Illustrates the sequential additive step-by-step contribution of each feature from the baseline expectation to the final output.



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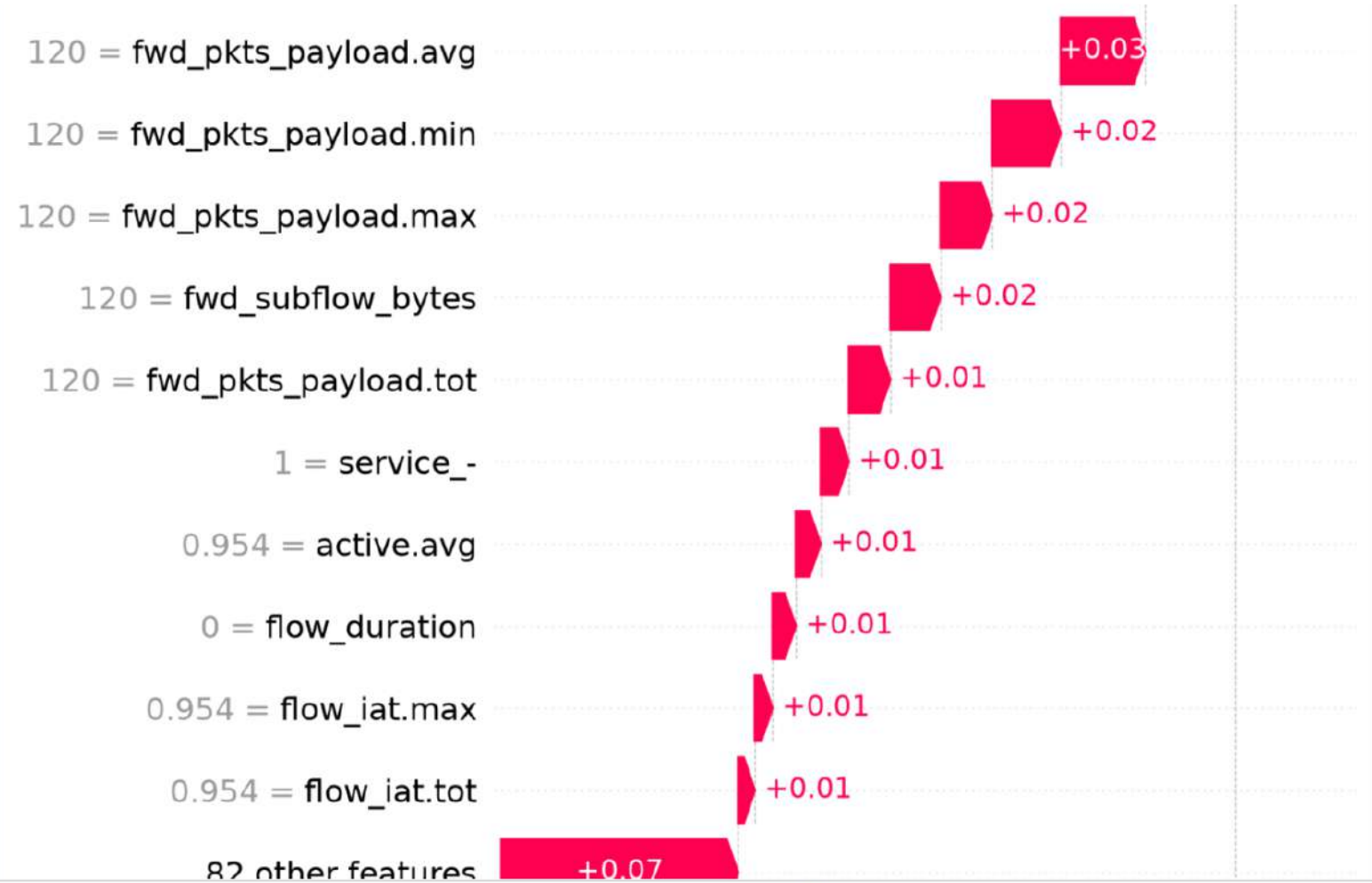
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Complete 93-Feature Telemetry Values & Signed Attributions

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CONTROLLED RESEARCH BENCHMARK

HELD-OUT TEST SET (N = 24,624)

Model Performance | Benchmark Evaluation

Rigorous technical evaluation of the 200-tree Random Forest classifier on the independent, held-out test partition. Examines classification accuracy, macro-averaged metrics, balanced accuracy, and category-level precision-recall trade-offs.

TEST ACCURACY	MACRO F1-SCORE	BALANCED ACCURACY	CORRECT FLOWS	TOTAL ERRORS
99.83%	96.22%	96.57%	24,581	43
0.998254 Exact	Treats all 12 classes equally	Mean class recall	99.825% of test data	0.175% error rate

1. Interactive 12-Class Confusion Matrix

Visualizes predicted classifications against verified ground-truth labels. Diagonal cells represent correct classifications; off-diagonal cells represent misclassifications.

Display Format:

- Raw Flow Counts
- Normalized (Recall % per True Class)

Held-Out Test Confusion Matrix (Raw Flow Counts)

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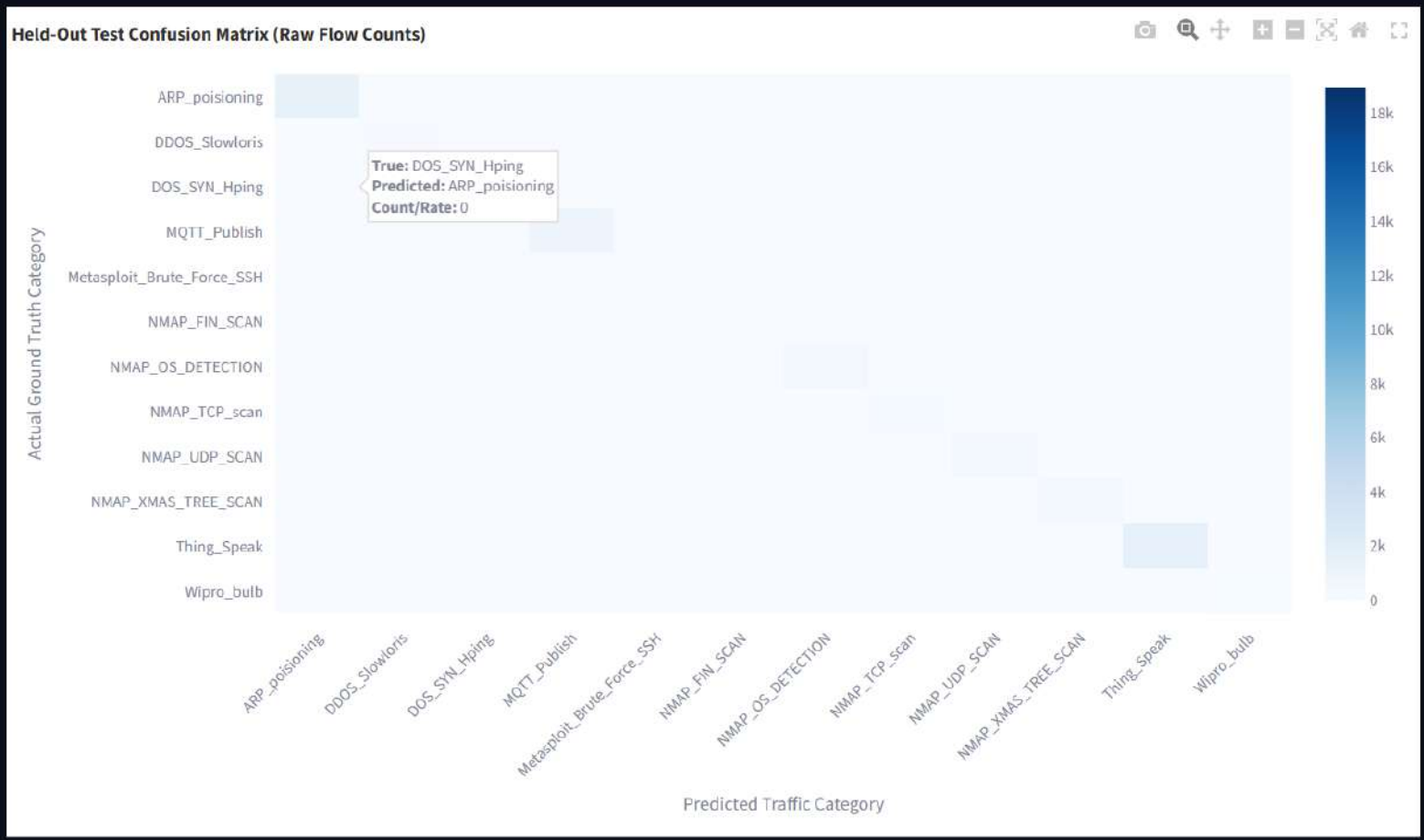
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Normalized (Recall % per True Class)



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2. Class-Wise Classification Performance

Breakdown of Precision, Recall, and F1-Score across all 12 traffic categories on the held-out test set.

⚠ **Imbalance & Low-Support Notice:** Certain rare attack categories have very small representation in the held-out test partition: **Metasploit_Brute_Force_SSH** (7 flows), **NMAP_FIN_SCAN** (6 flows). A single misclassification in these categories causes large shifts in recall percentages.

Traffic Category	Precision	Recall	F1-Score	Support
ARP_poisioning	98.72%	99.29%	99.00%	1,550
DDOS_Slowloris	98.11%	97.20%	97.65%	107
DOS_SYN_Hping	100.00%	100.00%	100.00%	18,932
MQTT_Publish	99.88%	99.76%	99.82%	829
Metasploit_Brute_Force_SSH	75.00%	85.71%	80.00%	7
NMAP_FIN_SCAN	83.33%	83.33%	83.33%	6
NMAP_OS_DETECTION	100.00%	100.00%	100.00%	400
NMAP_TCP_scan	100.00%	100.00%	100.00%	200
NMAP_UDP_SCAN	99.22%	98.84%	99.03%	518
NMAP_XMAS_TREE_SCAN	99.75%	99.50%	99.63%	402
Thing_Speak	99.32%	99.08%	99.20%	1,622
Wipro_bulb	98.00%	96.08%	97.03%	51

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Wipro_bulb	98.00%	96.08%	97.03%	51

3. Evaluation Methodology & Operational Considerations

Methodological Rigor

- **Strict Isolation:** The 24,624 held-out test observations were never exposed to model training.
- **Stratified Partitioning:** Preserved exact proportional representation of rare attack types across splits.
- **Unbiased Metrics:** Balanced accuracy (96.57%) and Macro F1 (96.22%) prevent majority-class dominance (such as DOS_SYN_Hping with 76.9% share) from inflating performance.

Real-World Operational Boundary

- **Benchmark Specificity:** High test accuracy reflects performance on the RT-IoT2022 test partition under controlled lab conditions.
- **Domain Shift:** Network telemetry in enterprise environments features novel protocols, encrypted tunnels, and noise not captured in lab datasets.
- **No Guarantee of Zero-Day Coverage:** Supervised models specialize in known patterns and require explainability for human oversight when novel anomalies occur.

Research Prototype Disclaimer: ShadowScan Phase 1 is an offline explainability research prototype evaluated on the benchmark **RT-IoT2022** dataset (93 features, 12 traffic classes, 200-tree Random Forest). It analyzes captured network flow telemetry records and does not perform live network packet capture, sniffing, or inline blocking. All reported metrics represent held-out benchmark test evaluations under controlled experimental conditions.

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SHAP TREEEXPLAINER | GLOBAL & CLASS ATTRIBUTION

Global XAI Insights | Explainability Governance

Dataset-level feature attributions derived from TreeExplainer. Evaluates which network telemetry metrics drive the classifier across all traffic, drills into class-specific signatures, and audits explanation stability.

PRIMARY GLOBAL DRIVER	SECONDARY GLOBAL DRIVER	VAL VS TEST STABILITY	GLOBAL TOP-10 OVERLAP
fwd_pkts_payload.avg	id.resp_p	0.9999	100.0%
Mean Abs SHAP: 0.0381	Destination Port (0.0336)	Spearman Rank Correlation	Identical Top-10 Feature Set

Global Feature Importance | Class-Wise Feature Profiles | Contribution Direction | Intra-Class Consistency | Validation vs. Test Stability

Top 20 Globally Important Features

Calculated by taking the average absolute SHAP value across all held-out test observations and output classes.



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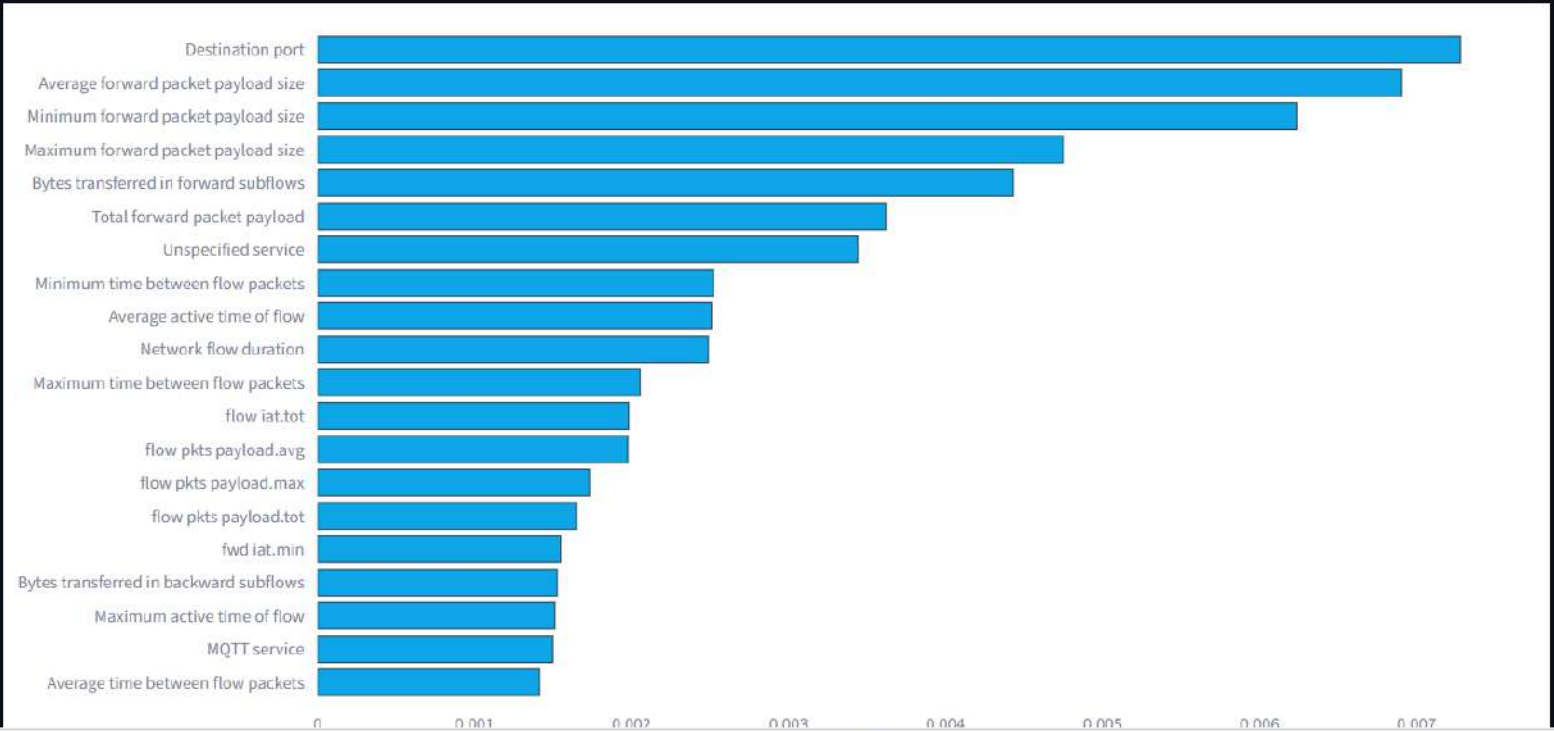
PHASE 1 - EXPLAINABLE NIDS

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Global Feature Importance | Class-Wise Feature Profiles | Contribution Direction | Intra-Class Consistency | Validation vs. Test Stability

Top 20 Globally Important Features

Calculated by taking the average absolute SHAP value across all held-out test observations and output classes.



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Global Feature Importance | Class-Wise Feature Profiles | Contribution Direction | Intra-Class Consistency | Validation vs. Test Stability

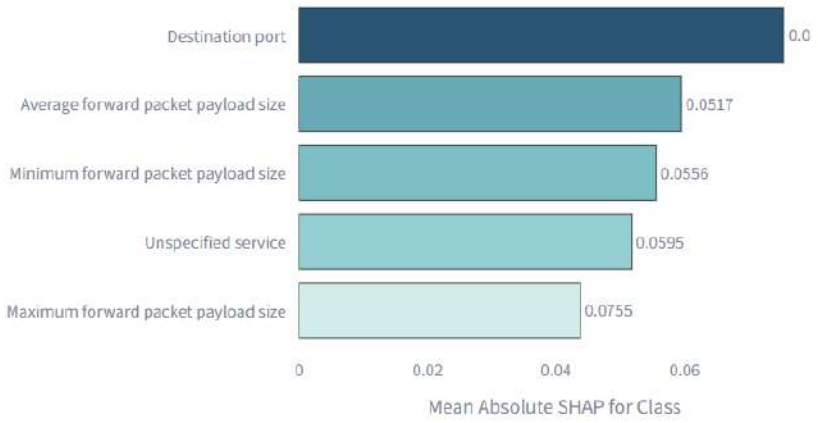
Class-Wise Feature Profiles (Top 5 Contributing Features)

Examine which specific features govern the model's recognition of individual attack categories.

Select Traffic Category:

ARP_poisioning

Top 5 Features: ARP_poisioning



Profile Table: ARP_poisioning

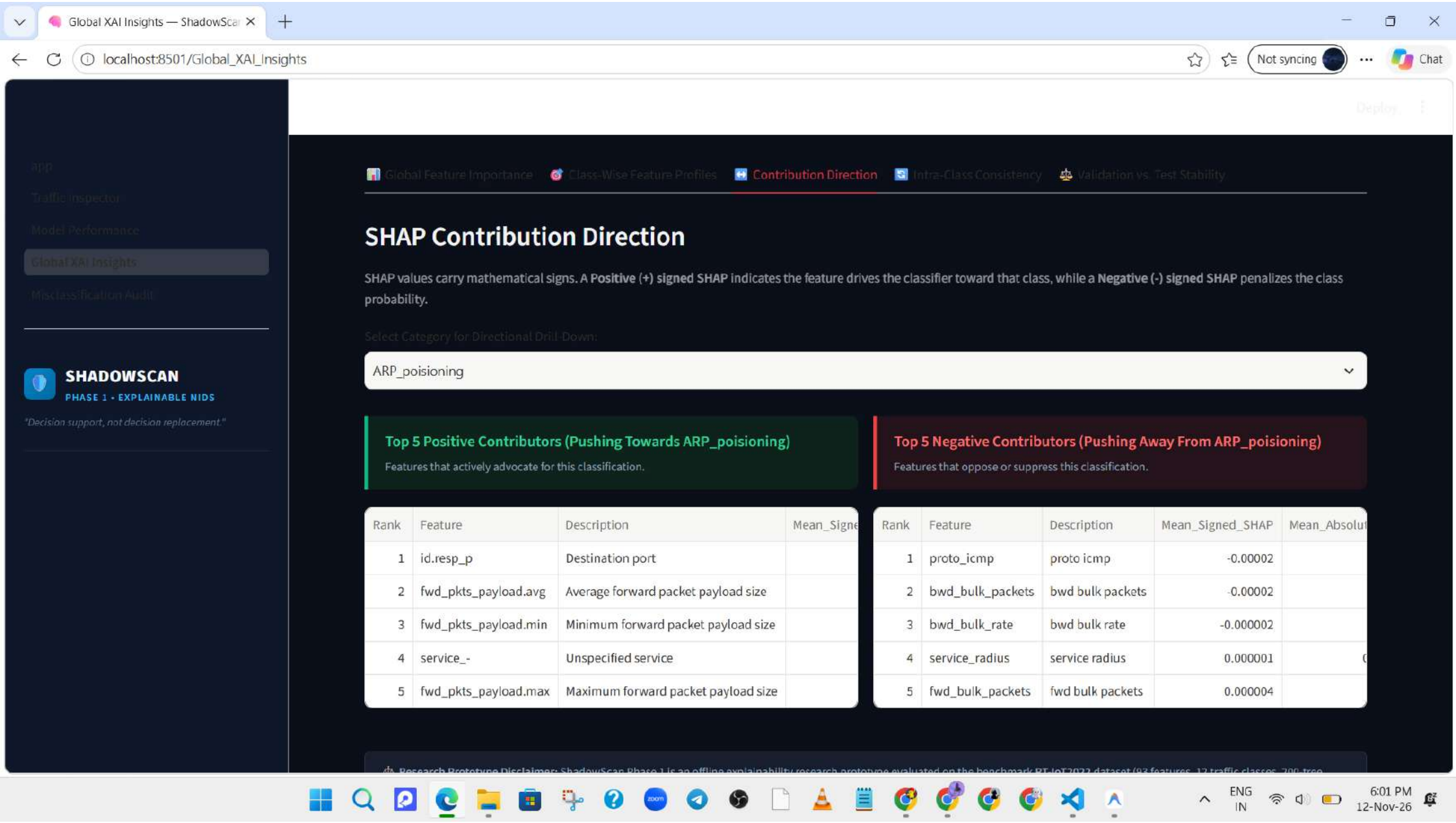
Test Set Support: 1550 observations

Rank	Feature	Description	Mea
1	id.resp_p	Destination port	
2	fwd_pkts_payload.avg	Average forward packet payload size	
3	fwd_pkts_payload.min	Minimum forward packet payload size	
4	service_-	Unspecified service	
5	fwd_pkts_payload.max	Maximum forward packet payload size	

Research Prototype Disclaimer: ShadowScan Phase 1 is an offline explainability research prototype evaluated on the benchmark **RT-IoT2022** dataset (93 features, 12 traffic classes, 200-tree

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Global Feature Importance
Class-Wise Feature Profiles
Contribution Direction
Intra-Class Consistency
Validation vs. Test Stability
SHAP Contribution Direction
SHAP values carry mathematical signs. A Positive (+) signed SHAP indicates the feature drives the classifier toward that class, while a Negative (-) signed SHAP penalizes the class probability.
Select Category for Directional Drill-Down:
ARP_poisioning
Top 5 Positive Contributors (Pushing Towards ARP_poisioning)
Features that actively advocate for this classification.
Top 5 Negative Contributors (Pushing Away From ARP_poisioning)
Features that oppose or suppress this classification.
Rank Feature Description Mean_Signe
1 id.resp_p Destination port
2 fwd_pkts_payload.avg Average forward packet payload size
3 fwd_pkts_payload.min Minimum forward packet payload size
4 service_- Unspecified service
5 fwd_pkts_payload.max Maximum forward packet payload size
Rank Feature Description Mean_Signed_SHAP Mean_Absolut
1 proto_icmp proto icmp -0.00002
2 bwd_bulk_packets bwd bulk packets -0.00002
3 bwd_bulk_rate bwd bulk rate -0.000002
4 service_radius service radius 0.000001
5 fwd_bulk_packets fwd bulk packets 0.000004
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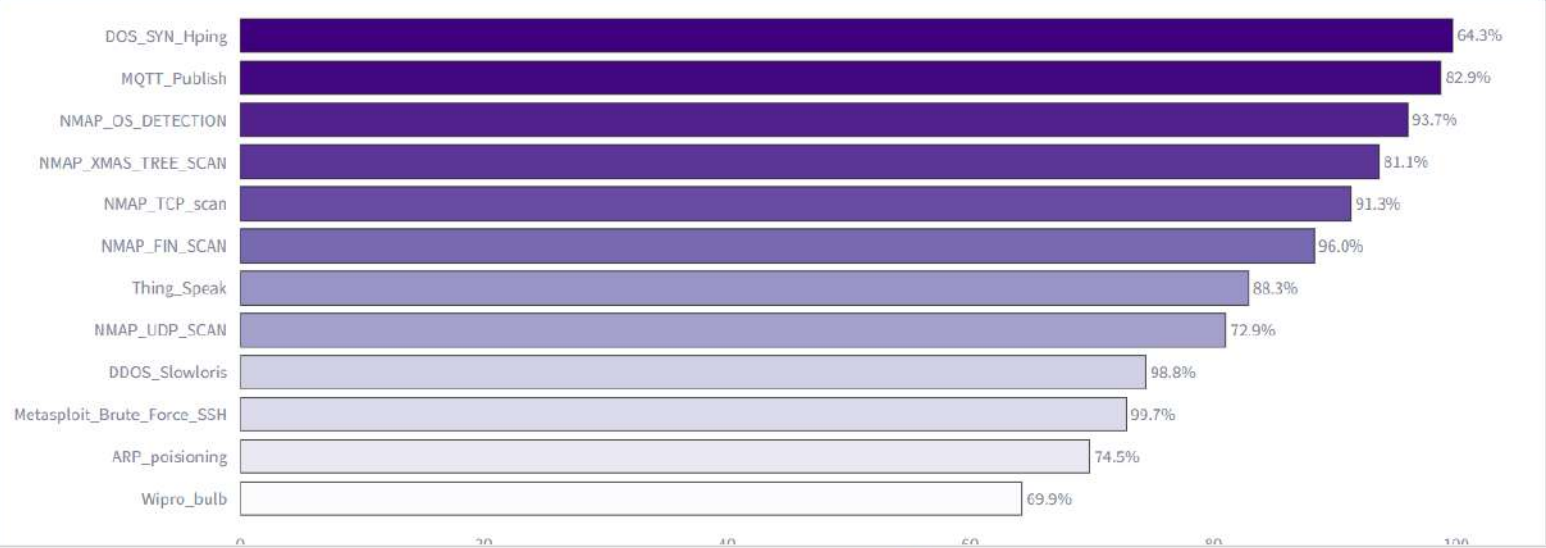
Global Feature Importance | Class-Wise Feature Profiles | Contribution Direction | Intra-Class Consistency | Validation vs. Test Stability

Intra-Class Explanation Consistency (Within-Class Homogeneity)

Metric Definition: Evaluated on the held-out test partition (from `shap_test_consistency.csv`). For each class *c* and each test flow belonging to *c*, we compute the percentage overlap between that individual flow's top-10 SHAP features and its class's aggregate top-10 SHAP features:

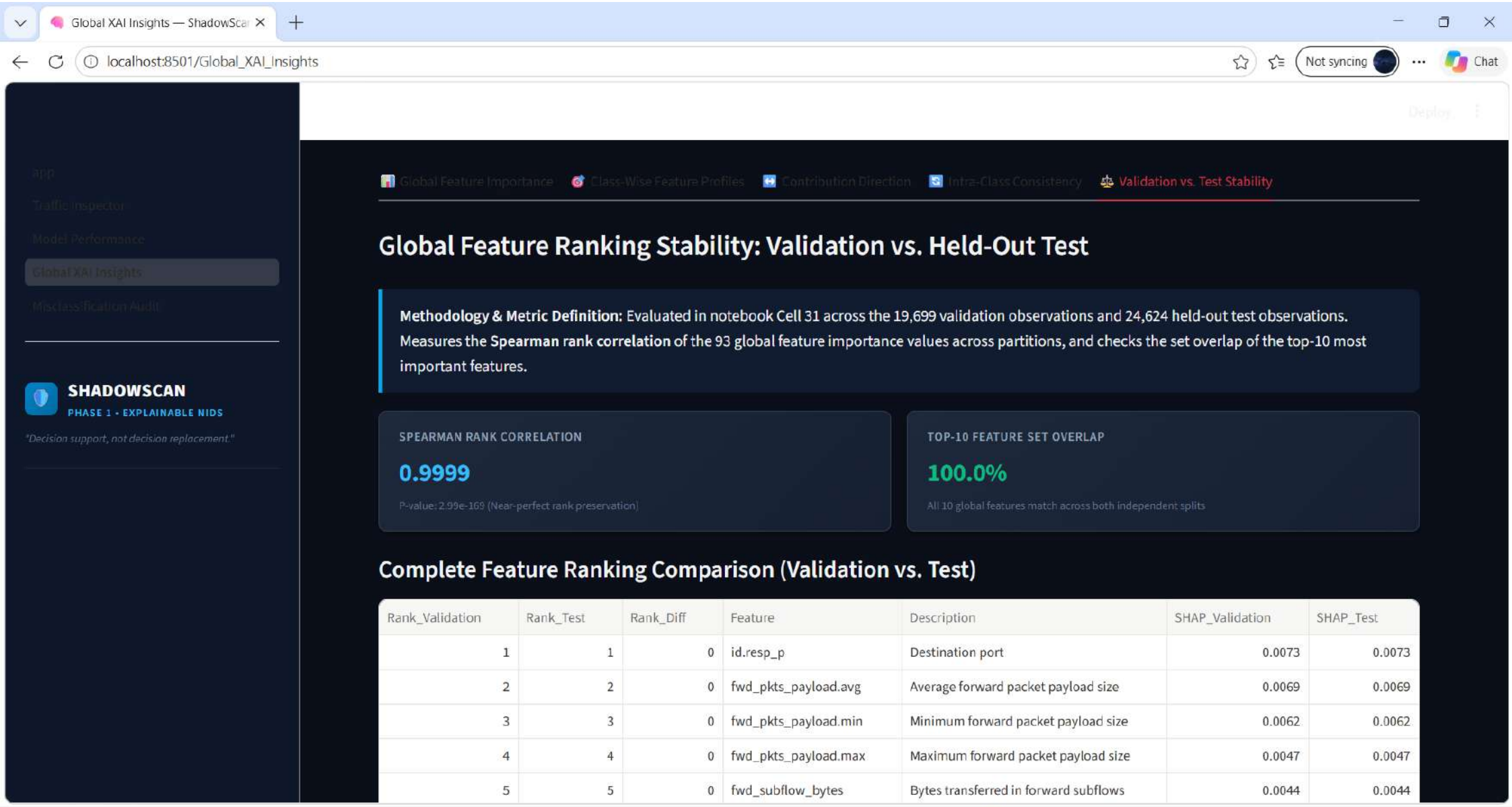
`Top-10 Overlap % = (|Individual_Top10 ∩ Class_Top10| / 10) * 100%`.

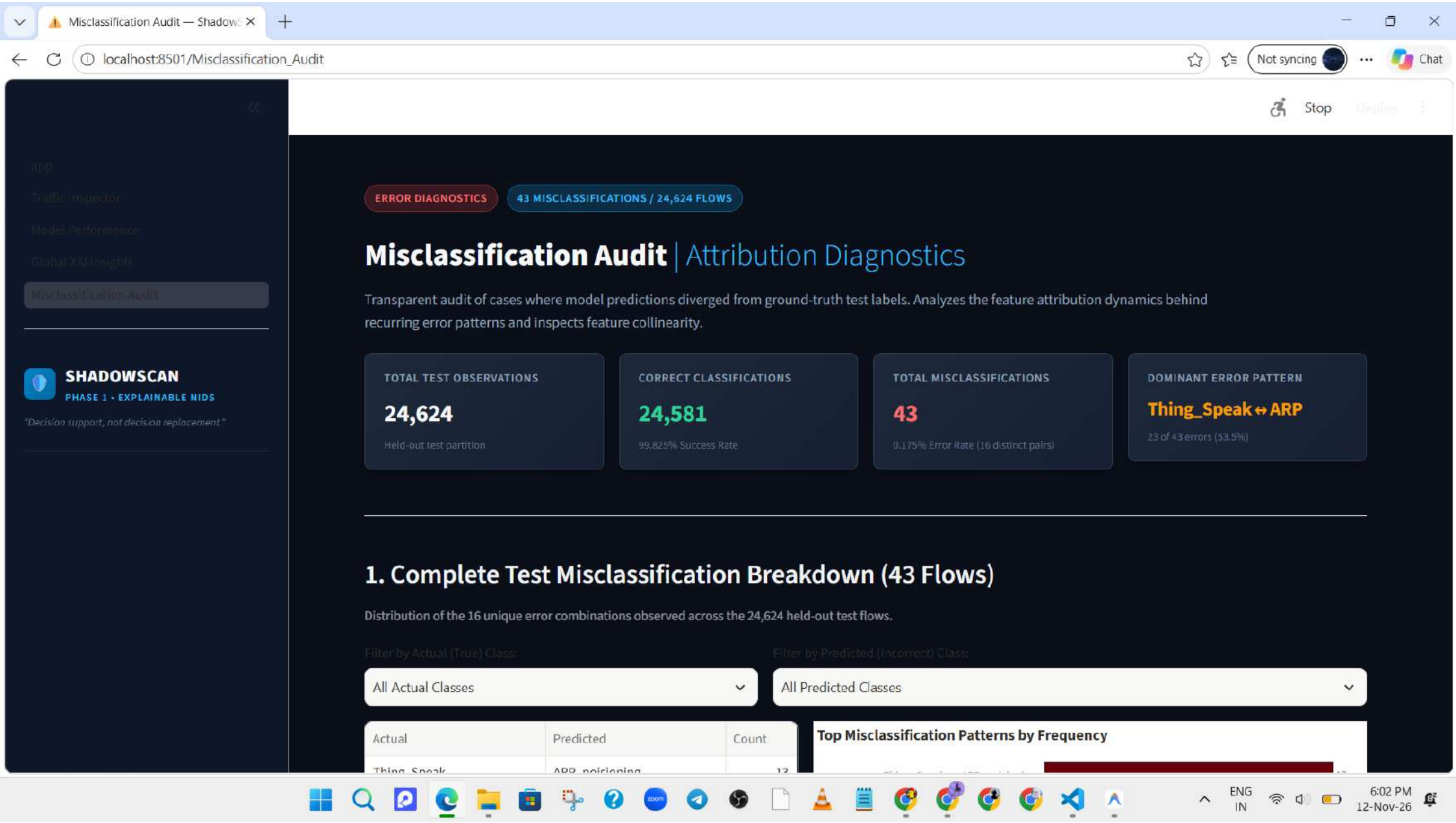
This measures **within-class explanation stability** — i.e., whether all flows of an attack type rely on the same core indicators.



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"Decision support, not decision replacement."
Global Feature Importance
Class-Wise Feature Profiles
Contribution Direction
Intra-Class Consistency
Validation vs. Test Stability
Global Feature Ranking Stability: Validation vs. Held-Out Test
Methodology & Metric Definition: Evaluated in notebook Cell 31 across the 19,699 validation observations and 24,624 held-out test observations. Measures the Spearman rank correlation of the 93 global feature importance values across partitions, and checks the set overlap of the top-10 most important features.
SPEARMAN RANK CORRELATION
0.9999
P-value: 2.99e-169 (Near-perfect rank preservation)
TOP-10 FEATURE SET OVERLAP
100.0%
All 10 global features match across both independent splits
Complete Feature Ranking Comparison (Validation vs. Test)
Rank_Validation | Rank_Test | Rank_Diff | Feature | Description | SHAP_Validation | SHAP_Test
1 | 1 | 0 | id.resp_p | Destination port | 0.0073 | 0.0073
2 | 2 | 0 | fwd_pkts_payload.avg | Average forward packet payload size | 0.0069 | 0.0069
3 | 3 | 0 | fwd_pkts_payload.min | Minimum forward packet payload size | 0.0062 | 0.0062
4 | 4 | 0 | fwd_pkts_payload.max | Maximum forward packet payload size | 0.0047 | 0.0047
5 | 5 | 0 | fwd_subflow_bytes | Bytes transferred in forward subflows | 0.0044 | 0.0044
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ERROR DIAGNOSTICS
43 MISCLASSIFICATIONS / 24,624 FLOWS
Misclassification Audit | Attribution Diagnostics
Transparent audit of cases where model predictions diverged from ground-truth test labels. Analyzes the feature attribution dynamics behind recurring error patterns and inspects feature collinearity.
TOTAL TEST OBSERVATIONS
24,624
Held-out test partition
CORRECT CLASSIFICATIONS
24,581
99.825% Success Rate
TOTAL MISCLASSIFICATIONS
43
0.175% Error Rate (16 distinct pairs)
DOMINANT ERROR PATTERN
Thing_Speak ↔ ARP
23 of 43 errors (53.5%)
1. Complete Test Misclassification Breakdown (43 Flows)
Distribution of the 16 unique error combinations observed across the 24,624 held-out test flows.
Filter by Actual (True) Class:
All Actual Classes
Filter by Predicted (Incorrect) Class:
All Predicted Classes
Actual
Predicted
Count
Top Misclassification Patterns by Frequency
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1. Complete Test Misclassification Breakdown (43 Flows)

Distribution of the 16 unique error combinations observed across the 24,624 held-out test flows.

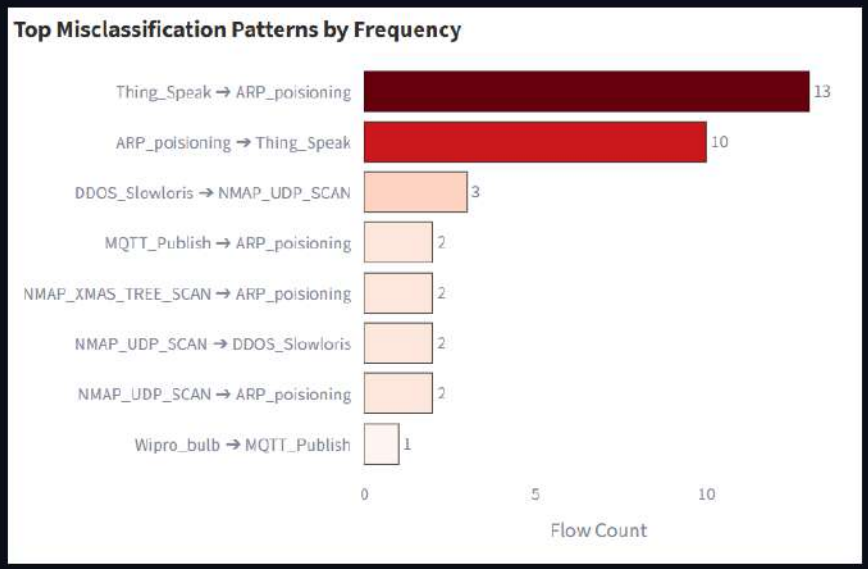
Filter by Actual (True) Class:

All Actual Classes

Filter by Predicted (Incorrect) Class:

All Predicted Classes

Actual	Predicted	Count
Thing_Speak	ARP_poisioning	13
ARP_poisioning	Thing_Speak	10
DDOS_Slowloris	NMAP_UDP_SCAN	3
NMAP_UDP_SCAN	ARP_poisioning	2
NMAP_UDP_SCAN	DDOS_Slowloris	2
NMAP_XMAS_TREE_SCAN	ARP_poisioning	2
MQTT_Publish	ARP_poisioning	2
Wipro_bulb	MQTT_Publish	1
NMAP_FIN_SCAN	NMAP_XMAS_TREE_SCAN	1
Wipro_bulb	Thing_Speak	1



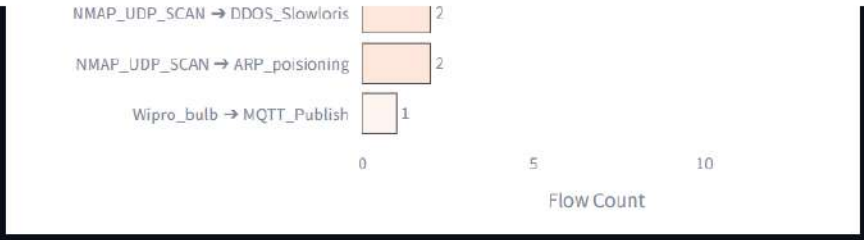
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Traffic Inspector
Model Performance
Global XAI Insights
Misclassification Audit

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PHASE 1 - EXPLAINABLE NIDS
"Decision support, not decision replacement."

NMAP_UDP_SCAN	Metasploit_Brute_Force_SSH	1
Metasploit_Brute_Force_SSH	ARP_poisoning	1
Thing_Speak	Wipro_bulb	1
NMAP_UDP_SCAN	NMAP_FIN_SCAN	1



2. Deep-Dive: Recurring Bidirectional Misclassifications

Key Research Finding: Over 53% of all test misclassifications (23 out of 43) occur between `Thing_Speak` (IoT telemetry) and `ARP_poisoning` (network spoofing).
The table and charts below compare the average SHAP contributions that **pulled the model toward the incorrect prediction** versus those that **advocated for the true ground-truth category.**

Select Recurring Error Group to Inspect:

Group 1: Actual = Thing_Speak → Predicted = ARP_poisoning (13 observations) Group 2: Actual = ARP_poisoning → Predicted = Thing_Speak (10 observations)

Features Pulling TOWARD Incorrect Prediction (ARP_poisoning)
Average positive SHAP contributions driving the false classification.

Features Advocating for TRUE Category (Thing_Speak)
Features that correctly opposed the error and pushed for the true ground truth.



Misclassification Audit — Shadow

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PHASE 1 - EXPLAINABLE NIDS

"Decision support, not decision replacement."

2. Deep-Dive: Recurring Bidirectional Misclassifications

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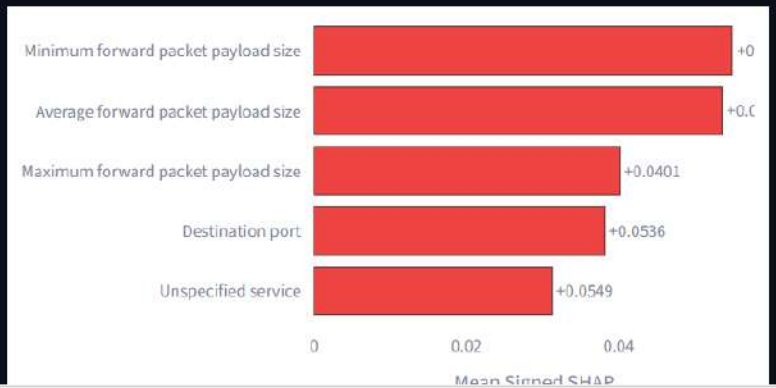
Select Recurring Error Group to Inspect:

Group 1: Actual = Thing_Speak → Predicted = ARP_poisioning (13 observations)

Group 2: Actual = ARP_poisioning → Predicted = Thing_Speak (10 observations)

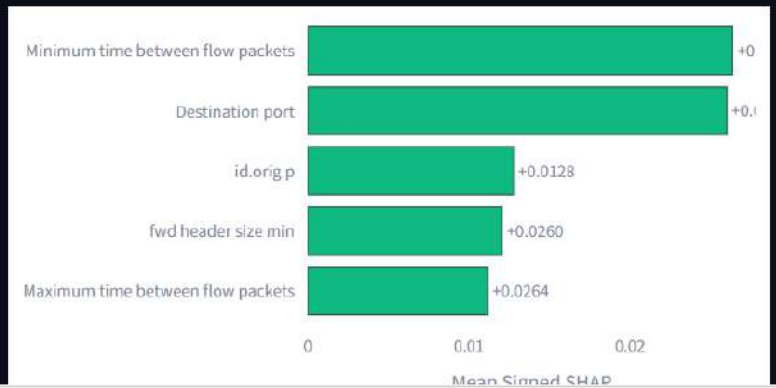
Features Pulling TOWARD Incorrect Prediction (ARP_poisioning)

Average positive SHAP contributions driving the false classification.



Features Advocating for TRUE Category (Thing_Speak)

Features that correctly opposed the error and pushed for the true ground truth.



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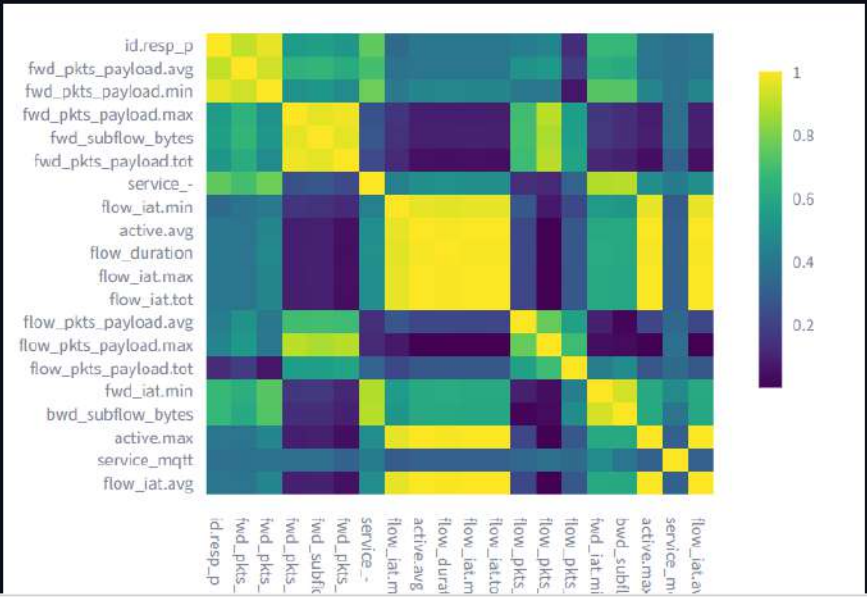
PHASE 1 - EXPLAINABLE NIDS

"Decision support, not decision replacement."

3. Feature Collinearity & Attribution Reliability Analysis

Methodological Caveat on Correlated Features: Tree-based models and additive attribution methods like SHAP can split or redistribute importance among highly collinear features. When two features are strongly correlated, high attribution on one feature does **not** imply the other is irrelevant. Below are feature pairs among the top-20 global features with absolute Spearman correlation `|ρ| ≥ 0.85`.

Spearman Rank Correlation Matrix (Top 20 Features)



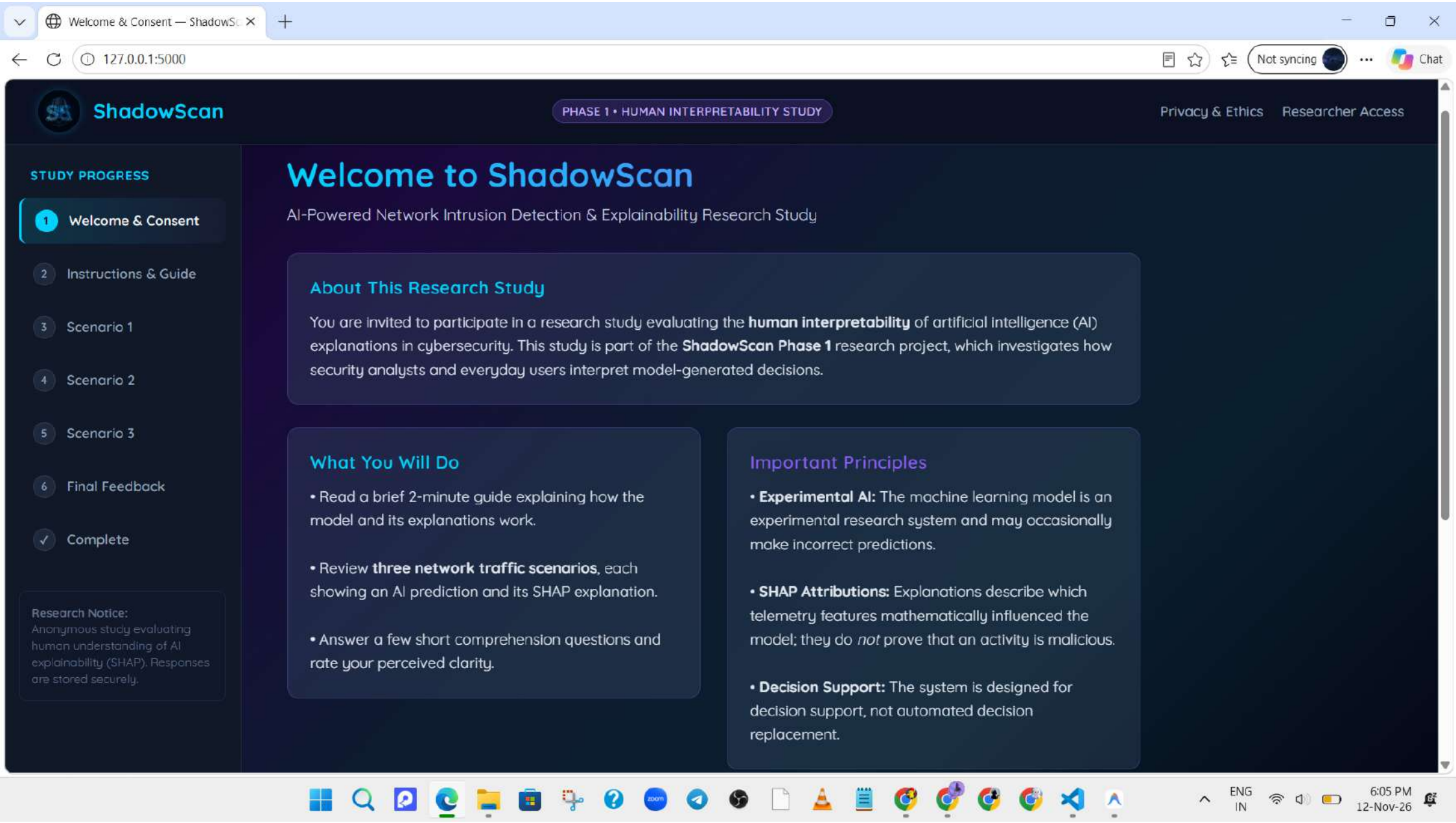
Strongly Correlated Pairs (|ρ| ≥ 0.85)

Feature pairs that may share or mask attribution during decision tree splits:

Feature 1	Feature 2	Spearman Correlation
flow_iat.max	flow_iat.tot	1
active.avg	active.max	0.9999
flow_iat.tot	flow_iat.avg	0.9999
flow_iat.max	flow_iat.avg	0.9999
active.avg	flow_iat.max	0.9986
flow_iat.max	active.max	0.9986
flow_iat.tot	active.max	0.9986
active.max	flow_iat.avg	0.9985
active.avg	flow_iat.tot	0.9985

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Research Notice:
Anonymous study evaluating human understanding of AI explainability (SHAP). Responses are stored securely.
Welcome to ShadowScan
AI-Powered Network Intrusion Detection & Explainability Research Study
About This Research Study
You are invited to participate in a research study evaluating the human interpretability of artificial intelligence (AI) explanations in cybersecurity. This study is part of the ShadowScan Phase 1 research project, which investigates how security analysts and everyday users interpret model-generated decisions.
What You Will Do
• Read a brief 2-minute guide explaining how the model and its explanations work.
• Review three network traffic scenarios, each showing an AI prediction and its SHAP explanation.
• Answer a few short comprehension questions and rate your perceived clarity.
Important Principles
• Experimental AI: The machine learning model is an experimental research system and may occasionally make incorrect predictions.
• SHAP Attributions: Explanations describe which telemetry features mathematically influenced the model; they do not prove that an activity is malicious.
• Decision Support: The system is designed for decision support, not automated decision replacement.
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Study Instructions & Guide

A quick 2-minute overview of how to interpret the model predictions and explainability bars.

1. What does the AI Model do?

In modern computer networks, devices exchange telemetry flows (data packets, ports, transmission rates, and durations). The ShadowScan **Random Forest classifier** examines these telemetry features to predict whether traffic resembles normal operation or a specific category of network attack.

2. What are SHAP Explanations?

Machine learning models are often called "black boxes" because their internal calculations are complex. **SHAP (SHapley Additive exPlanations)** breaks down the model's prediction by measuring how much each individual feature pushed the decision higher or lower.

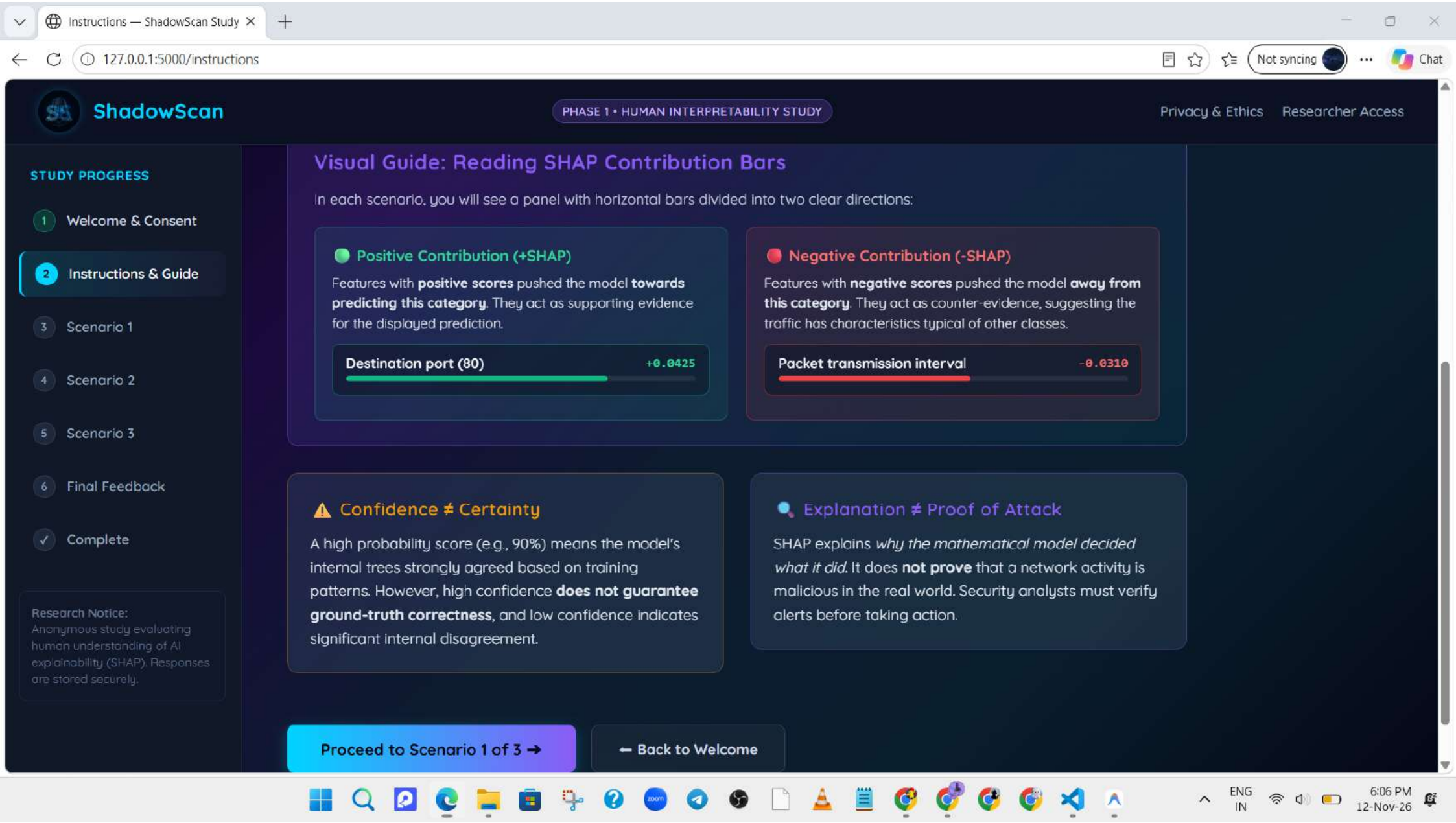
Visual Guide: Reading SHAP Contribution Bars

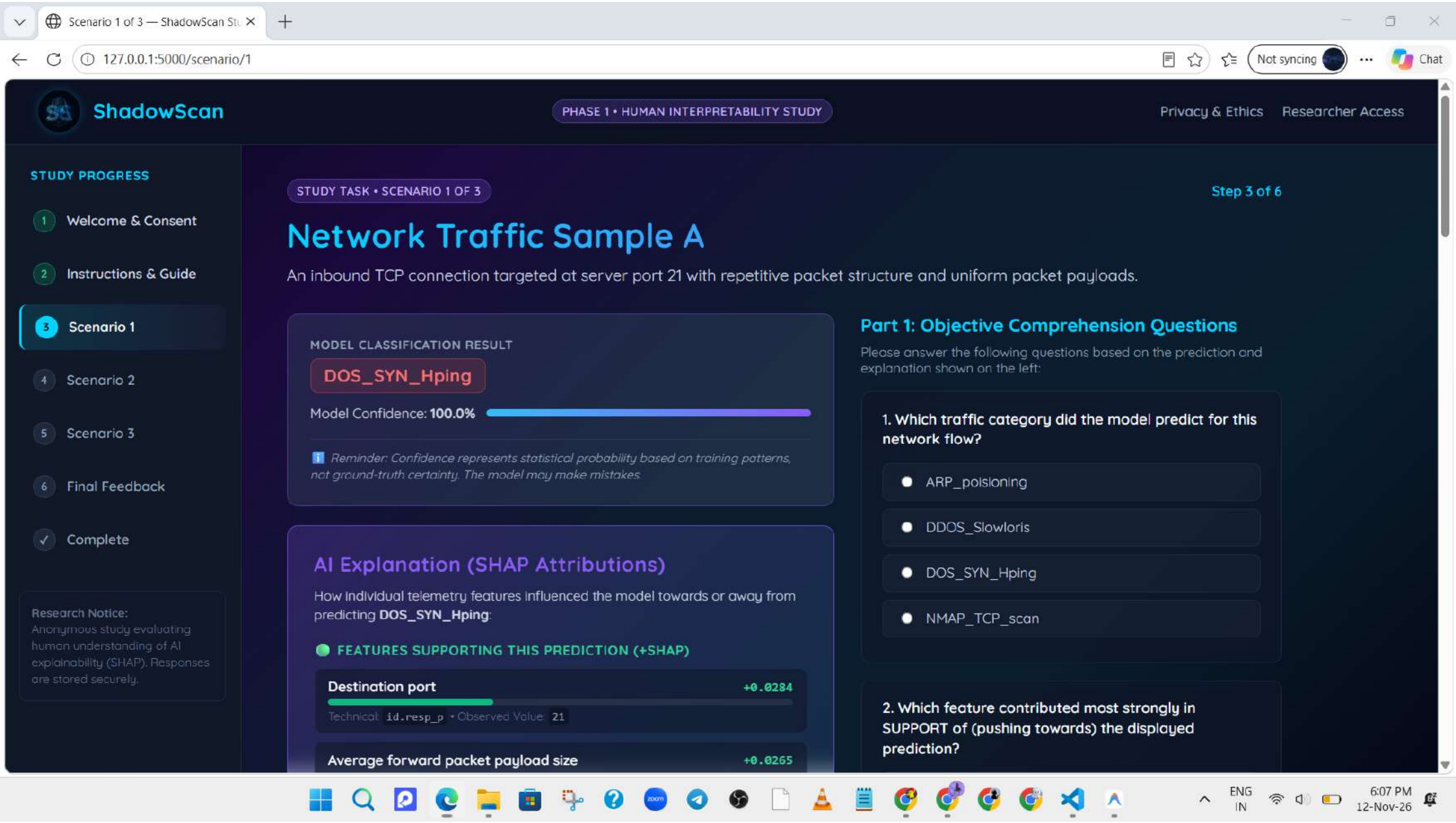
In each scenario, you will see a panel with horizontal bars divided into two clear directions:

Positive Contribution (+SHAP)

Negative Contribution (-SHAP)

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Research Notice:
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Visual Guide: Reading SHAP Contribution Bars
In each scenario, you will see a panel with horizontal bars divided into two clear directions:
Positive Contribution (+SHAP)
Features with positive scores pushed the model towards predicting this category. They act as supporting evidence for the displayed prediction.
Destination port (80) +0.0425
Negative Contribution (-SHAP)
Features with negative scores pushed the model away from this category. They act as counter-evidence, suggesting the traffic has characteristics typical of other classes.
Packet transmission interval -0.0310
Confidence ≠ Certainty
A high probability score (e.g., 90%) means the model's internal trees strongly agreed based on training patterns. However, high confidence does not guarantee ground-truth correctness, and low confidence indicates significant internal disagreement.
Explanation ≠ Proof of Attack
SHAP explains why the mathematical model decided what it did. It does not prove that a network activity is malicious in the real world. Security analysts must verify alerts before taking action.
Proceed to Scenario 1 of 3 →
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STUDY TASK • SCENARIO 1 OF 3
Step 3 of 6
Network Traffic Sample A
An inbound TCP connection targeted at server port 21 with repetitive packet structure and uniform packet payloads.
MODEL CLASSIFICATION RESULT
DOS_SYN_Hping
Model Confidence: 100.0%
Reminder: Confidence represents statistical probability based on training patterns, not ground-truth certainty. The model may make mistakes.
AI Explanation (SHAP Attributions)
How individual telemetry features influenced the model towards or away from predicting DOS_SYN_Hping:
FEATURES SUPPORTING THIS PREDICTION (+SHAP)
Destination port +0.0284
Technical: id.resp_p • Observed Value: 21
Average forward packet payload size +0.0265
Part 1: Objective Comprehension Questions
Please answer the following questions based on the prediction and explanation shown on the left:
1. Which traffic category did the model predict for this network flow?
ARP_poisioning
DDOS_Slowloris
DOS_SYN_Hping
NMAP_TCP_scan
2. Which feature contributed most strongly in SUPPORT of (pushing towards) the displayed prediction?
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AI Explanation (SHAP Attributions)

How individual telemetry features influenced the model towards or away from predicting **DOS_SYN_Hping**:

FEATURES SUPPORTING THIS PREDICTION (+SHAP)

Destination port +0.0284
Technical: `id.resp_p` • Observed Value: `21`

Average forward packet payload size +0.0265
Technical: `fwd_pkts_payload.avg` • Observed Value: `120`

Minimum forward packet payload size +0.0216
Technical: `fwd_pkts_payload.min` • Observed Value: `120`

Maximum forward packet payload size +0.0163
Technical: `fwd_pkts_payload.max` • Observed Value: `120`

Bytes transferred in forward subflows +0.0158
Technical: `fwd_subflow_bytes` • Observed Value: `120`

FEATURES OPPOSING THIS PREDICTION (-SHAP)

Source port -0.0003
Technical: `id.orig_p` • Observed Value: `5.499e+04`

- DOS_SYN_Hping
- NMAP_TCP_scan

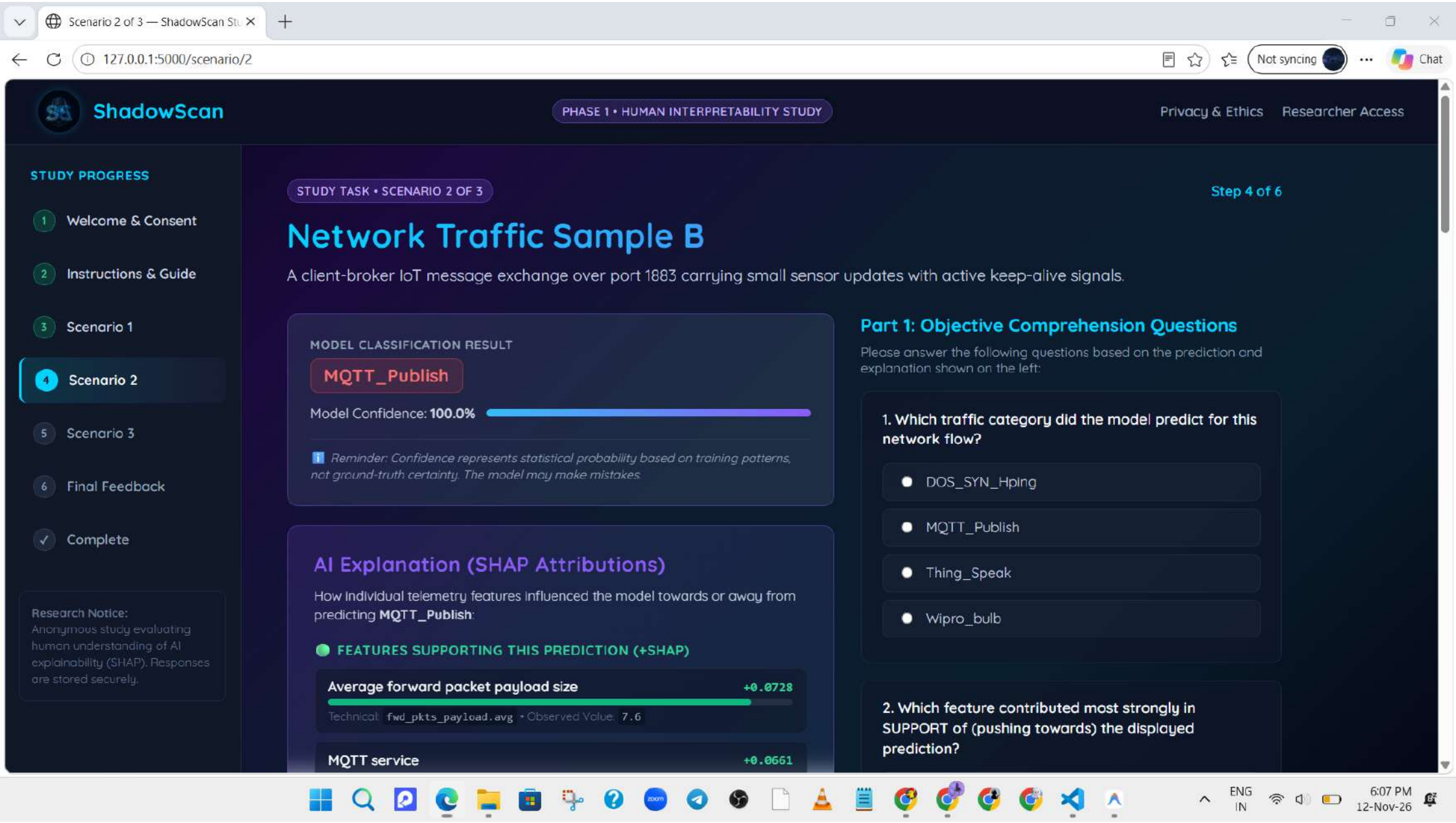
2. Which feature contributed most strongly in SUPPORT of (pushing towards) the displayed prediction?

- Average forward packet payload size
- Destination port
- Source port

3. Which feature contributed most strongly AGAINST (opposing / pushing away from) the displayed prediction?

- Average forward packet payload size
- Destination port
- Source port

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STUDY TASK • SCENARIO 2 OF 3
Step 4 of 6
Network Traffic Sample B
A client-broker IoT message exchange over port 1883 carrying small sensor updates with active keep-alive signals.
MODEL CLASSIFICATION RESULT
MQTT_Publish
Model Confidence: 100.0%
Reminder: Confidence represents statistical probability based on training patterns, not ground-truth certainty. The model may make mistakes.
AI Explanation (SHAP Attributions)
How individual telemetry features influenced the model towards or away from predicting MQTT_Publish:
FEATURES SUPPORTING THIS PREDICTION (+SHAP)
Average forward packet payload size +0.0728
Technical: fwd_pkts_payload.avg • Observed Value: 7.6
MQTT service +0.0661
Part 1: Objective Comprehension Questions
Please answer the following questions based on the prediction and explanation shown on the left:
1. Which traffic category did the model predict for this network flow?
DOS_SYN_Hping
MQTT_Publish
Thing_Speak
Wipro_bulb
2. Which feature contributed most strongly in SUPPORT of (pushing towards) the displayed prediction?
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How individual telemetry features influenced the model towards or away from predicting **MQTT_Publish**:

FEATURES SUPPORTING THIS PREDICTION (+SHAP)

Average forward packet payload size +0.0728
Technical: `fwd_pkts_payload.avg` • Observed Value: `7.6`

MQTT service +0.0661
Technical: `service_mqtt` • Observed Value: `1`

Destination port +0.0493
Technical: `id.resp_p` • Observed Value: `1883`

Average active time of flow +0.0455
Technical: `active.avg` • Observed Value: `2.247e+06`

Bytes transferred in forward subflows +0.0431
Technical: `fwd_subflow_bytes` • Observed Value: `25.33`

FEATURES OPPOSING THIS PREDICTION (-SHAP)

Initial forward TCP window size -0.0007
Technical: `fwd_init_window_size` • Observed Value: `6.424e+04`

flow FIN flag count -0.0000
Technical: `flow_FIN_flag_count` • Observed Value: `0`

- Wipro_bulb

2. Which feature contributed most strongly in SUPPORT of (pushing towards) the displayed prediction?

- Average forward packet payload size
- Initial forward TCP window size
- MQTT service
- flow FIN flag count

3. Which feature contributed most strongly AGAINST (opposing / pushing away from) the displayed prediction?

- Average forward packet payload size
- Initial forward TCP window size
- MQTT service

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Bytes transferred in forward subflows +0.0431

Technical: `fwd_subflow_bytes` • Observed Value: `25.33`

FEATURES OPPOSING THIS PREDICTION (-SHAP)

Initial forward TCP window size -0.0007

Technical: `fwd_init_window_size` • Observed Value: `6.424e+04`

flow FIN flag count -0.0000

Technical: `flow_FIN_flag_count` • Observed Value: `0`

idle.tot -0.0000

Technical: `idle.tot` • Observed Value: `5.981e+07`

active.std -0.0000

Technical: `active.std` • Observed Value: `0`

bwd bulk packets -0.0000

Technical: `bwd_bulk_packets` • Observed Value: `0`

*SHAP values quantify feature contributions to the mathematical decision, they do not prove causation or confirm real-world maliciousness.

3. Which feature contributed most strongly AGAINST (opposing / pushing away from) the displayed prediction?

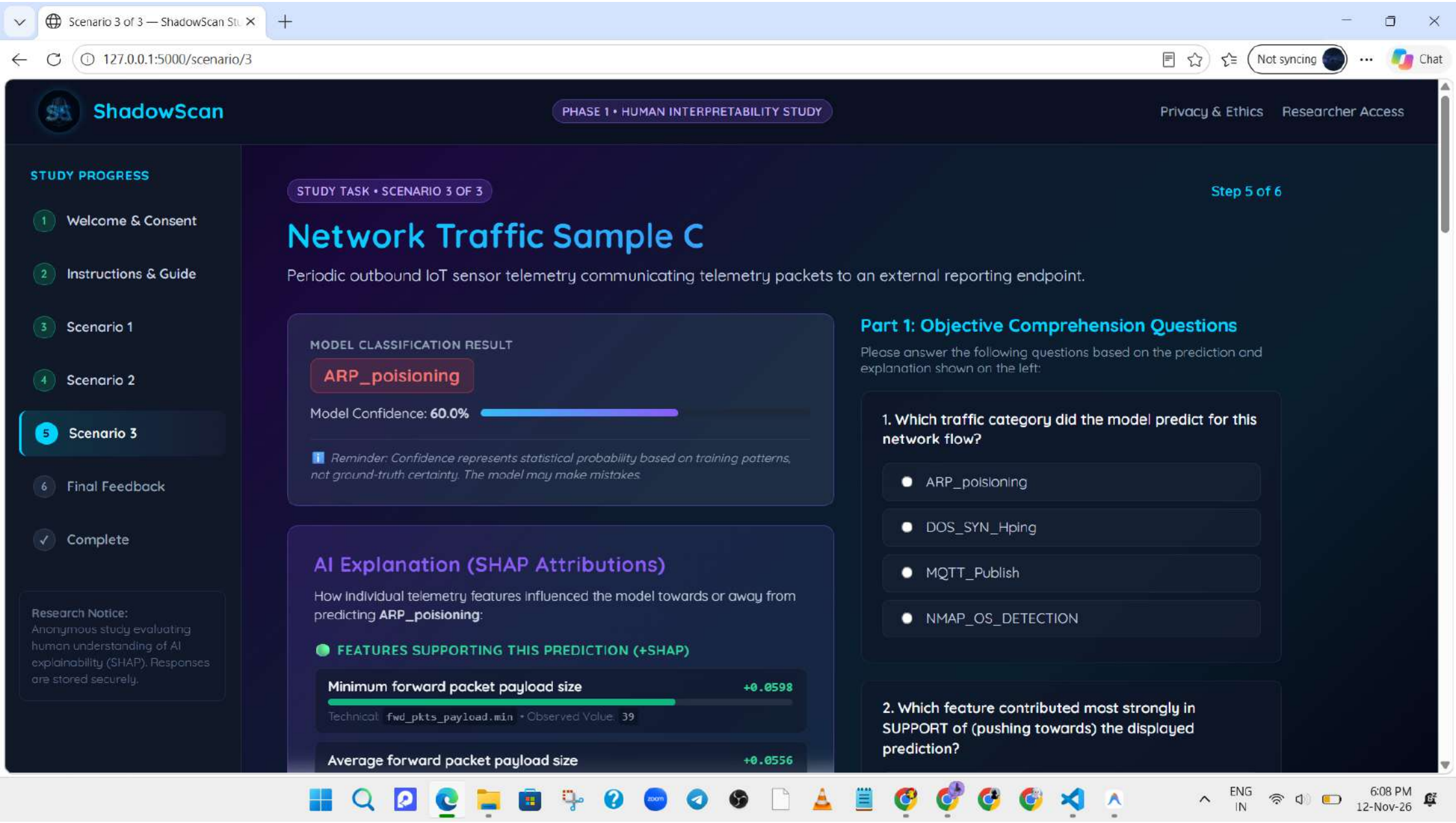
- Average forward packet payload size
- Initial forward TCP window size
- MQTT service
- flow FIN flag count

4. In this explanation, what does a positive (+) SHAP contribution indicate?

- It pushed the model towards predicting the displayed category.
- It proved that the network flow was a confirmed malicious attack.
- It pushed the model away from predicting the displayed category.

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Research Notice:
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STUDY TASK • SCENARIO 3 OF 3
Step 5 of 6
Network Traffic Sample C
Periodic outbound IoT sensor telemetry communicating telemetry packets to an external reporting endpoint.
MODEL CLASSIFICATION RESULT
ARP_poisioning
Model Confidence: 60.0%
Reminder: Confidence represents statistical probability based on training patterns, not ground-truth certainty. The model may make mistakes.
AI Explanation (SHAP Attributions)
How individual telemetry features influenced the model towards or away from predicting ARP_poisioning:
FEATURES SUPPORTING THIS PREDICTION (+SHAP)
Minimum forward packet payload size +0.0598
Technical: fwd_pkts_payload.min • Observed Value: 39
Average forward packet payload size +0.0556
Part 1: Objective Comprehension Questions
Please answer the following questions based on the prediction and explanation shown on the left:
1. Which traffic category did the model predict for this network flow?
ARP_poisioning
DOS_SYN_Hping
MQTT_Publish
NMAP_OS_DETECTION
2. Which feature contributed most strongly in SUPPORT of (pushing towards) the displayed prediction?
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predicting **ARP_poisioning**:

FEATURES SUPPORTING THIS PREDICTION (+SHAP)

Minimum forward packet payload size	+0.0598	Technical: `fwd_pkts_payload.min` • Observed Value: `39`
Average forward packet payload size	+0.0556	Technical: `fwd_pkts_payload.avg` • Observed Value: `39`
Destination port	+0.0443	Technical: `id.resp_p` • Observed Value: `53`
Unspecified service	+0.0370	Technical: `service_-` • Observed Value: `0`
Maximum forward packet payload size	+0.0335	Technical: `fwd_pkts_payload.max` • Observed Value: `39`

FEATURES OPPOSING THIS PREDICTION (-SHAP)

Minimum time between flow packets	-0.0344	Technical: `flow_iat.min` • Observed Value: `8.543e+04`
Source port	-0.0121	Technical: `id.orig_p` • Observed Value: `5.301e+04`

- NMAP_OS_DETECTION

2. Which feature contributed most strongly in SUPPORT of (pushing towards) the displayed prediction?

- Average forward packet payload size
- Minimum forward packet payload size
- Minimum time between flow packets
- Source port

3. Which feature contributed most strongly AGAINST (opposing / pushing away from) the displayed prediction?

- Average forward packet payload size
- Minimum forward packet payload size
- Minimum time between flow packets

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Unspecified service +0.0370
Technical: `service_-` • Observed Value: `0`

Maximum forward packet payload size +0.0335
Technical: `fwd_pkts_payload.max` • Observed Value: `39`

FEATURES OPPOSING THIS PREDICTION (-SHAP)

Minimum time between flow packets -0.0344
Technical: `flow_iat.min` • Observed Value: `8.543e+04`

Source port -0.0121
Technical: `id.orig_p` • Observed Value: `5.301e+04`

Standard deviation of backward packet payload -0.0103
Technical: `bwd_pkts_payload.std` • Observed Value: `0`

flow iat.std -0.0049
Technical: `flow_iat.std` • Observed Value: `0`

bwd subflow pkts -0.0030
Technical: `bwd_subflow_pkts` • Observed Value: `1`

** SHAP values quantify feature contributions to the mathematical decision; they do not prove causation or confirm real-world maliciousness.*

Start

- Source port

3. Which feature contributed most strongly AGAINST (opposing / pushing away from) the displayed prediction?

- Average forward packet payload size
- Minimum forward packet payload size
- Minimum time between flow packets
- Source port

4. In this explanation, what does a positive (+) SHAP contribution indicate?

- It pushed the model towards predicting the displayed category.
- It proved that the network flow was a confirmed malicious attack.
- It pushed the model away from predicting the

Study Completed — ShadowScan

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Thank You for Participating!

Your responses have been recorded anonymously and saved to the study database.

ANONYMOUS ID: 91384F89-136F-4C06-BF90-39F662A40B9F

Research Debriefing & Context

In this study, you evaluated how AI explainability (SHAP) helps humans understand model decisions for network intrusion detection.

About the Scenarios You Inspected:

- **Scenario 1 & 2** demonstrated instances where the model correctly identified high-confidence attack and IoT protocols.
- **Scenario 3** represented a subtle, real-world machine learning error: an IoT sensor stream (**Thing_Speak**) was misclassified as **ARP_poisioning**. Notice how the opposing features in that scenario specifically pointed to transmission intervals typical of sensor data. Evaluating whether users can detect such contradictory evidence in explanations is a core goal of this research.

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